Cayden J. Osborne

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Education

M.A. Mathematics *August 2026 - May 2027*
University at Albany (SUNY), Albany, NY *(Expected)*
Thesis: A Proof that Thompson's Group F Is Of Type F_∞
Advisor: Dr. Matthew Zaremsky

B.S. Mathematics (Physics Minor) *August 2023 - Dec. 2026*
University at Albany (SUNY), Albany, NY *(Expected)*
Major GPA: 3.71/4.00 Overall GPA: 3.56/4.00

Research Interests

Algebraic Number Theory | Geometric Group Theory | Algebra | Topology

Research Experience & Projects

Graduate Independent Study *Planned*
Department of Mathematics & Statistics, University at Albany, SUNY
Supervisor: Dr. Matthew Zaremsky

- Planned study of Thompson groups, finiteness properties, and the meaning of type F_∞.
- Will examine CW complexes and group actions as tools for connecting algebraic and topological properties of groups.

UAlbany Showcase Day Presenter April 2026
Department of Mathematics & Statistics, University at Albany, SUNY
Supervisor: Dr. Anupam Srivastav

- Studied Gaussian Primes and their structure in the ring $\mathbb{Z}[i]$, including all of the characterizations by using norms, and unique factorization. Presented findings at the University-Wide research showcase.

Undergraduate Independent Study August 2025
Department of Mathematics & Statistics, University at Albany, SUNY
Supervisor: Dr. Changlong Zhong

- Studied fundamental properties of groups, including subgroups, cosets, and quotient groups.
- Explored homomorphisms, isomorphisms, and the isomorphism theorems.

Teaching Experience

- **Substitute Teacher**, Mathematics and Physics 2024 - Present
  Instructed classes including AP Calculus, AP Physics, Geometry, Algebra, and Statistics.
- **Private Tutor**, Mathematics and Physics 2022 – Present
  Tutored high school mathematics and physics courses, and undergraduate calculus

Coursework

Graduate: Number Theory, Theory of Statistics I, Advanced Linear Algebra, Topology I

Advanced Undergraduate: Abstract Algebra, Probability Theory, Statistics, Computational Mathematics

Relevant Physics: Newtonian Mechanics, Electromagnetism, Modern Physics, Waves and Optics, Computational Physics, and Neutrino Astronomy

Honors & Awards

- Dean's List, University at Albany, SUNY, 2024-2026
- William Roth Scholarship, University at Albany, SUNY, 2025-2027

Skills

Mathematical Software: LaTeX, Python (NumPy, SciPy)
Programming: Python, Java, C/C++
Languages: English (Native), Spanish (intermediate)